MANUAL SENIBINA PROMPT MULTI-ENGINE

SOP & BLUEPRINT EKSEKUSI UGC AVATAR AI (GOOGLE FLOW, NANO BANANA & IMAGEN)

Dokumen ini merupakan panduan teknikal standard operasi bagi penghasilan aset video kempen UGC (User-Generated Content) menggunakan kecerdasan buatan pelbagai enjin. Kandungan dokumen ini wajib dipatuhi oleh operator media, content strategist, dan QA auditor bagi menjamin ketekalan visual, kualiti tipografi, dan pencegahan halusinasi dimensi produk.

1. MATRIKS STRATEGI ENJIN & PENJAJARAN MOD

Penghasilan video pemasaran digital berskala besar memerlukan pemilihan mod operasi yang tepat mengikut ketersediaan aset rujukan. Jadual di bawah menetapkan fungsi, kelebihan, dan risiko utama bagi setiap mod penjanaan:

MOD OPERASI	KAEDAH UTAMA (METHOD)	KELEBIHAN TEKNIKAL	RISIKO KRITIKAL
Text-to-Video (T2V)	Generasi sifar berdasarkan deskripsi parameter linguistik universal.	Kebebasan kreatif mutlak, tiada pergantungan kepada photoshoot luaran.	Herotan struktur (Structural Drift) & kegagalan teks jenama.
Frames (F2V)	Mengunci bingkai asal ($t=0$) sebagai sauh piksel statik.	Ketekalan identiti avatar dan geometri produk 1:1 pada saat awal.	Mutasi skala produk (Enlargement Hallucination) semasa bergerak.
Ingredients	Dekomposisi elemen semantik (Wajah, Objek, Gaya) secara berasingan.	Kebebasan kinematik ekstrem; membolehkan perubahan sudut kamera dinamik.	Kebocoran Tekstur (Texture Bleeding) antara tangan dan produk.
Image Guidance	Modifikasi setempat menggunakan kaedah 'Delta Instruction'.	Pengekalan subjek tegar semasa melakukan pertukaran latar belakang/pakaian.	Ralat pembiasan cahaya pada sempadan geometri mikro.

2. PANDUAN STRUKTUR PROMPT VIDEO (GOOGLE FLOW)

A. Mod Text-to-Video (T2V)

Method: Membina keseluruhan babak daripada matriks rawak (latent noise) melalui pengekodan teks berlapis. Wajib mengisytiharkan geometrik rigid secara literal untuk mengehadkan kebebasan enjin.

Structure Format:

```
[SUBJECT_INITIALIZATION] -> [PRODUCT_AND_SPATIAL_SPECIFICATIONS] -> [UGC_PERFORMANCE_VECTORS] ->
[ENVIRONMENT_AND_CINEMATOGRAPHY] -> [TEMPORAL_PHYSICS_CONSTRAINTS]
```

CONTOH PROMPT LENGKAP T2V:

```
[SUBJECT_INITIALIZATION]
A professional Southeast Asian (Malay) woman in her early 30s, clean complexion, tied-back dark
hair, wearing casual modest corporate smart clothing. She is the onscreen UGC creator, starting
the scene with an approachable, confident smile.

[PRODUCT_AND_SPATIAL_SPECIFICATIONS]
She is holding a rigid, cylindrical 10ml micro-product dropper bottle. The bottle properties:
dark amber-tinted glass, matte white paper label wrapped around the center, with clean black
typographic markings. The avatar holds the 10ml micro-bottle using a strict, stable pinch grip
at the base. To prevent scaling hallucinations, enforce a precise 2mm visual air gap between her
fingertips and the main printed logo on the label. The product size must remain proportional to
a human hand, strictly maintaining a 1:4 product-to-hand ratio.

[UGC_PERFORMANCE_VECTORS]
The woman delivers an authentic product demonstration, looking directly into the camera lens
with unbreaking eye contact. Realistic jaw and lip movements synchronized with an expressive
speaking cadence. Natural, frequent micro-expressions: subtle head tilts, authentic blinking
patterns, and light nodding to emphasize points.

[ENVIRONMENT_AND_CINEMATOGRAPHY]
Shot on a simulated 4K smartphone front camera, 28mm wide lens, medium close-up framing. The
setting is a brightly lit, modern minimalist office space with soft diffused daylight entering
from a side window. Warm skin tones, natural skin textures with visible pores, no synthetic
airbrushing. The camera features a realistic, low-amplitude handheld micro-shake to mimic
authentic user-generated content.

[TEMPORAL_PHYSICS_CONSTRAINTS]
Enforce absolute structural integrity on the product bottle: straight lines must remain
straight, labels must not morph, dissolve, or swim across frames. Hands and all five fingers
must maintain proper anatomical constraints and joint placements during movement. Zero frame
distortion, fluid 30fps motion vectors, complete temporal consistency from start to finish.
```

B. Mod Frames (F2V / Frame-to-Video)

Method: Menetapkan imej rujukan sebagai input bingkai permulaan. Teks prompt bertindak sebagai vektor perubahan temporal (motion vector delta) tanpa membina semula data identiti asal.

Structure Format:

```
[IMAGE_REF_ANCHOR] -> [PERFORMANCE_DYNAMICS] -> [SPATIAL_INTERACTION_AND_PRODUCT_LOCK] ->
[CINEMATOGRAPHY_AND_ENVIRONMENT] -> [PHYSICS_CONSTRAINTS]
```

CONTOH PROMPT LENGKAP F2V:

```
[IMAGE_REF: ANCHOR_START]
Maintaining strict 1:1 identity, features, clothing, and background from the uploaded reference
frame.

[PERFORMANCE_DYNAMICS]
The female avatar in the reference image transitions into active speaking motion, delivering a
natural, authentic UGC-style product review. Direct and continuous eye-contact with the camera
lens. Realistic facial muscle movement, subtle eyebrow raises, and micro-expressions of
enthusiasm. Natural mouth shapes and lip-sync cadence matching an energetic speaking rhythm.

[SPATIAL_INTERACTION_AND_PRODUCT_LOCK]
The avatar maintains a precise pinch grip on the 10ml micro-product bottle, ensuring a visible
2mm air gap between the holding fingers and the main text of the product label to lock physical
scale. The product geometry, scale, and text branding must remain 100% rigid and chemically
stable, completely free from warping, expansion, or texture swimming as the hand moves. Natural,
micro-hand gestures congruent with standard human conversation.

[CINEMATOGRAPHY_AND_ENVIRONMENT]
Subtle, high-frequency handheld camera micro-shake to simulate authentic smartphone UGC
recording. Lens properties: 24mm close-up profile. Maintain the soft, warm studio lighting from
the reference image, with realistic, dynamic micro-shadows shifting across the avatar's fingers
and face as she performs small, natural movements.

[PHYSICS_CONSTRAINTS]
Zero frame-warping. Absolute anatomical preservation of hands, fingers, and facial structure.
Temporal consistency locked across all generated frames. No morphing of the product outline.
```

C. Mod Ingredients

Method: Meleraikan imej rujukan kepada komponen bebas. Enjin mengekstrak data identiti entiti secara berasingan dan membina semula komposisi mengikut geometri kamera baharu.

Structure Format:

```
[INGREDIENTS_EXTRACTION] -> [NEW_COMPOSITION_AND_STAGING] -> [INTERACTION_AND_SPATIAL_MATH] ->
[DIGITAL_RENDERING_CONSTRAINTS]
```

CONTOH PROMPT LENGKAP MOD INGREDIENTS:

```
[INGREDIENTS_EXTRACTION]
- INGREDIENT_A (Subject Identity): Extract exclusively the facial features, skin tone,
hairstyle, and biological identity of the female avatar from the reference image.
- INGREDIENT_B (Product Identity): Extract the rigid cylindrical geometry, dark amber glass
material texture, and white label typography of the 10ml micro-product bottle.
- DISCARD: Original background and static camera framing from the reference image.

[NEW_COMPOSITION_AND_STAGING]
Recompose the extracted ingredients into a completely new cinematic sequence. The camera starts
at a low-angle close-up of the product, then executes a smooth, stabilized upward vertical pan
to reveal the avatar’s face. The avatar is positioned in a modern, bright, minimalist vanity
room with natural sunlight filtering through soft linen curtains in the background.

[INTERACTION_AND_SPATIAL_MATH]
The extracted Ingredient_A (avatar's hand) interacts directly with Ingredient_B (the product).
The hand must execute a flawless, anatomically correct pinch grip near the base of the 10ml
bottle. Maintain a strict 2mm air gap between the fingertips and the printed text on the label.
The boundary between the organic skin texture of the fingers and the rigid glass/paper texture
of the product must remain perfectly sharp, with zero texture bleeding, zero warping, and zero
edge-smudging.

[DIGITAL_RENDERING_CONSTRAINTS]
Enforce rigid-body physics on Ingredient_B; its dimensions, straight lines, and branding layout
must remain immutable throughout the camera pan. Dynamic shadows must project accurately from
the bottle onto the avatar's palm based on the new sunlight direction. 30fps temporal
synchronization, zero ghosting artifacts.
```

3. PANDUAN STRUKTUR PROMPT IMEJ (NANO BANANA & IMAGEN 3)

Method: Penjanaan berpandukan imej rujukan (Reference-Guided Mode) menggunakan fungsi Arahan Delta untuk melakukan pengeditan lokal tanpa menjejaskan integriti struktur global.

Structure Format:

```
[REFERENCE_IMAGE_LOCK] -> [LOCAL_EDIT_DELTA] -> [TYPOGRAPHY_AND_BRANDING_LOCK] ->
[SPATIAL_MATH_AND_PROPORTIONS] -> [OUTPUT_SPECIFICATION]
```

CONTOH PROMPT LENGKAP PENJANAAN IMEJ (NANO BANANA 2/PRO & IMAGEN 3):

```
[REFERENCE_IMAGE_LOCK]
Using the uploaded reference image as the absolute foundation for subject identity. Preserve
100% of the facial structure, facial features, and natural skin tone of the female avatar.
Maintain the exact physical cylindrical shape and material properties of the product bottle.

[LOCAL_EDIT_DELTA]
Modify only the clothing and the background environment. Change the avatar's outfit from the
casual shirt in the reference image to a crisp, professional pastel-colored corporate blouse.
Replace the entire original background with a modern, high-end minimalist skincare studio
setting. Soft natural daylight filtering through large windows, creating a premium commercial
atmosphere.

[TYPOGRAPHY_AND_BRANDING_LOCK]
The text and brand markings on the product label must remain perfectly legible, crisp, and high-
fidelity. Enforce exact typographic preservation from the reference image with zero character
morphing or text distortion.

[SPATIAL_MATH_AND_PROPORTIONS]
Maintain a strict 1:4 product-to-hand ratio for the 10ml micro-cylinder bottle to prevent
enlargement hallucinations. The fingers holding the bottle must utilize a precise pinch grip at
the base, leaving a clear 2mm visual air gap over the printed brand logo. The boundary between
the organic skin lines of the fingers and the glass texture of the bottle must remain perfectly
sharp and distinct.

[OUTPUT_SPECIFICATION]
Studio-quality commercial photography, shallow depth of field, sharp focus on the avatar's eyes
and the product label, 4K resolution fidelity, clean digital rendering without synthetic
artifacts.
```

AMARAN OPERASI TEKNIKAL

Enjin Nano Banana Pro mempunyai sensitiviti tinggi terhadap ralat ruang (whitespace error). Sebarang trailing whitespaces atau ketidakseragaman sintaks nod pada fail konfigurasi luaran (seperti skrip automasi RPA) boleh mencetuskan status 'Fetch Deadlock'. Pastikan sintaks disunting bersih sebelum eksekusi render.

4. STANDARD OPERATING PROCEDURE (SOP) PENJANAAN DI GOOGLE FLOW

Langkah 1: Audit Aset Permulaan (Fasa Pra-Penjanaan)

- **1.1 Verification:** Semak kualiti imej rujukan. Resolusi minimum wajib bersaiz 1080p dengan pencahayaan jelas pada label produk.
- **1.2 Separation Audit:** Pastikan jari avatar tidak menutup lebih daripada 15% bahagian teks utama pada label produk di dalam imej rujukan.

Langkah 2: Penyediaan Konfigurasi & Pemilihan Mod

- **2.1 Mod F2V:** Digunakan jika pergerakan avatar adalah minimal (hanya bercakap tanpa memusingkan botol).
- **2.2 Mod Ingredients:** Digunakan jika perancangan video memerlukan perubahan sudut pandang (e.g., dari produk ke wajah atau pergerakan kanta dramatik).
- **2.3 Penguncian Model:** Gunakan tetapan parameter kawalan `image_guidance_scale` pada julat 0.75 - 0.85 untuk menjamin ketekalan geometri produk.

Langkah 3: Pemasangan Prompt Bertapis (Prompt Assembly)

- **3.1** Salin struktur blueprint mengikut mod yang dipilih daripada dokumen ini.

- **3.2** Ubah suai bahagian entiti demografi dan persekitaran mengikut keperluan kempen spesifik, namun *wajib mengekalkan blok matematik ruang* (2mm air gap dan nisbah 1:4).

Langkah 4: Pemantauan Proses Render (Fasa Eksekusi)

- **4.1** Laksanakan render ujian (Test Render) sepanjang 3 saat (90 bingkai pada kadar 30fps) terlebih dahulu sebelum melakukan komitmen render penuh.
- **4.2** Perhatikan sebarang tanda 'Texture Swimming' pada saat kedua penjanaan.

Langkah 5: Kawalan Kualiti (Fasa Pasca-Penjanaan)

- **5.1 Audit Anatomi:** Semak struktur tangan avatar pada setiap 30 bingkai. Pastikan tiada pertambahan atau pengurangan bilangan jari secara halusinasi.
- **5.2 Audit Tipografi:** Hentikan (Pause) video pada saat akhir dan lakukan zum 200% pada label produk untuk mengesahkan teks tidak bertukar menjadi aksara rosak.

5. CHECKLIST PRA-RENDERS & FAILSAFE MANUAL

Sila gunakan senarai semak di bawah sebelum menekan butang eksekusi render bagi mengelakkan pembaziran kredit token komersial:

NO	PERKARA YANG DIPERIKSA (AUDIT POINT)	STATUS STANDARD	TINDAKAN JIKA GAGAL (FAILSAFE ACTION)
1	Pemberatan Panduan Imej (Guidance Scale)	Berada pada julat 0.75 - 0.85	Laraskan slider manual ke atas; jangan biarkan pada tetapan auto/default (0.50).
2	Ketetapan Matematik Ruang (Spatial Math)	Frasa '2mm air gap' & '1:4 ratio' wujud dalam prompt	Suntik semula perenggan [SPATIAL_MATH] ke dalam prompt utama.
3	Sintaks Struktur Prompt	Tiada trailing whitespaces atau ruang kosong di akhir baris	Gunakan pembersih teks (text stripper) sebelum menampal skrip ke konsol Google Flow.
4	Kestabilan Sempadan Objek (Edge Smudging)	Sempadan kulit tangan dan botol produk tajam	Aktifkan arahan 'rigid-body physics constraint' pada sub-blok prompt terakhir.
5	Pengekalan Teks Label (Typography Clear)	Teks jenama tidak mencair/ swimming selepas saat pertama	Tukarkan operasi dari Mod T2V kepada Mod F2V atau naikkan nilai `frame_influence` ke 0.90.